

# Literature Review: Pose-Based Motion Embeddings and Contextual Metric Learning for Robust Retrieval

Giacomo Cappelletto – Research Project with Prof. Brian Kulis

July 6, 2026

## Abstract

The literature does not suggest that contextual metric learning has already been applied to noisy pose-sequence retrieval. Instead, it provides strong but mostly separate components: Kulis’s contextual similarity objective for image retrieval, public skeleton/motion encoders, one-shot skeleton metric-learning baselines, robust skeleton-recognition work, and motion-language retrieval systems. The most feasible first result is therefore not a new architecture, but a controlled adaptation: use a public NTU skeleton pipeline, expose a normalized sequence embedding, train standard metric-learning baselines and the contextual loss under the same balanced batches, and compare retrieval degradation under synthetic pose corruptions. The recommended first implementation is MotionBERT’s NTU120 one-shot embedding path plus the contextual-loss repository; MMAction2/PoseC3D is the stronger second-stage robustness baseline.

## 1 Project Framing

Prof. Kulis’s research agenda explicitly includes metric learning, content-based search, clustering, online learning, and applications to audio/visual data (Kulis, nd). The proposed project should therefore be framed as a metric-learning and retrieval problem first:

$$\begin{aligned} \text{pose sequence} &\rightarrow \text{sequence encoder} \rightarrow \ell_2\text{-normalized embedding} \\ &\rightarrow \text{nearest-neighbor retrieval} \rightarrow \text{ranking metrics under clean/noisy poses.} \end{aligned} \tag{1}$$

The central question is narrow enough to test:

Does neighborhood-based contextual similarity preserve pose-sequence retrieval quality better than standard pairwise or contrastive objectives when input skeletons are corrupted?

The relevant corruptions are joint jitter, missing joints, limb dropout, temporal frame dropout, camera/view perturbations, left-right swaps where anatomically meaningful, and 2D-to-3D lifting or detector confidence noise. Classification accuracy alone is not the target. A classifier can remain strong while its penultimate embedding is poorly ranked for retrieval. The correct evaluation unit is a query/gallery retrieval protocol with Recall@K, mAP, mAP@R, and robustness curves.

The main literature gap is the intersection of three lines of work: contextual metric learning for retrieval (Liao et al., 2023); skeleton/motion encoders for action recognition or representation learning (Yan et al., 2018; Duan et al., 2022; Zhu et al., 2023); and robustness studies for incomplete or noisy skeletons (Song et al., 2020; Shi et al., 2020). I found strong adjacent work, but not a paper in the supplied archive or subsequent source checks that combines contextual loss, temporal pose embeddings, retrieval ranking metrics, and a controlled pose-noise benchmark.

## 2 Contextual Metric Learning

Liao et al. (2023) propose supervised metric learning for image retrieval via contextual similarity optimization. Standard metric-learning losses operate mainly on pairwise or tuplewise distances; their method additionally asks whether two samples have similar neighborhoods. In the public implementation, embeddings are normalized, a batch-level cosine similarity matrix is formed, top- $k$  neighborhood masks are computed, and a contextual Jaccard-like similarity is matched to the label agreement matrix (Liao, nd).

The important transfer point is that the loss is mostly encoder-agnostic. For pose sequences, the image CNN can be replaced with a TCN, ST-GCN, PoseC3D, MotionBERT, or a simple GRU/Transformer encoder, while preserving the loss-side contract:

- the model returns one fixed-length embedding per sequence;
- embeddings are  $\ell_2$  normalized so dot products are cosine similarities;
- each training batch contains multiple examples per class;
- the contextual neighborhood size  $k$  is tied to the samples-per-class value  $M$ ;
- contextual loss is compared against contrastive, triplet, SupCon, multi-similarity, and classification-derived embeddings.

The repository audit surfaced one concrete engineering caveat. The 224-by-224 branch exposes the hybrid loss as `HybridLoss(..., k=args.IPC, eps=args.eps)`, while the 256-by-256 path hard-codes  $k = 4$  inside its contextual loss. For this project, the loss should be refactored so  $k = M$  is always passed explicitly; otherwise the batch sampler and contextual neighborhood definition can silently diverge.

## 3 What Has Been Done Before

Table 1 separates adjacent literatures by what they actually solve. This distinction matters because several papers produce useful embeddings, but only a subset evaluate retrieval directly, and fewer still evaluate retrieval under noisy pose inputs.

The strongest prior-art warning is from Musgrave et al. (2020a): metric-learning gains can disappear under careful controls. A defensible result should therefore report clean and noisy retrieval for every loss on the same data split, backbone, augmentation policy, batch composition, embedding dimension, and evaluator.

## 4 Datasets and Retrieval Targets

The first benchmark should be easy enough that the research contribution is the loss and robustness evaluation, not dataset plumbing. Table 2 ranks the relevant datasets accordingly.

The recommendation is to use NTU120 if data preparation is already available, or NTU60 as a pilot if MotionBERT/MMAAction2 setup time dominates. SportsPose and FineGYM should be treated as validation of relevance after the core methodology works.

## 5 Repository Audit

The practical build question is where contextual loss can be inserted with minimal new code. The audit covered the contextual-loss implementation, MotionBERT, MMAAction2, standalone ST-

Table 1: Relevant literature clusters and project implications.

| Cluster                     |          | Representative work                                                                                                                                                                                                                                                                      | Implication for this project                                                                                                                                                                                                           |
|-----------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Metric learning and ranking |          | Contrastive loss (Hadsell et al., 2006), classification baselines (Zhai and Wu, 2018), SupCon (Khosla et al., 2020), Smooth-AP (Brown et al., 2020), multi-similarity (Wang et al., 2019), PML (Musgrave et al., 2020b), and the metric-learning reality check (Musgrave et al., 2020a). | These define fair baselines. Contextual loss should not be compared only to cross-entropy; it should be compared to strong retrieval objectives on the same backbone and sampler.                                                      |
| Pose embeddings             |          | Pose Embeddings (Mori et al., 2015); VIPE/Pr-VIPE and occlusion-robust Pr-VIPE (Sun et al., 2019; Liu et al., 2020b).                                                                                                                                                                    | This line is closest conceptually: pose similarity is learned as an embedding problem. Pr-VIPE explicitly targets cross-view pose retrieval and partial keypoints, but it is not a supervised action-class motion-retrieval benchmark. |
| Skeleton recognition        | action   | ST-GCN (Yan et al., 2018), MST-GCN (Lee et al., 2022), PoseC3D (Duan et al., 2022), MotionBERT (Zhu et al., 2023).                                                                                                                                                                       | These provide encoders. Most optimize top-1 action accuracy, so their features need retrieval evaluation and metric-loss fine-tuning.                                                                                                  |
| Robust recognition          | skeleton | RA-GCN and PR-GCN (Song et al., 2020; Shi et al., 2020).                                                                                                                                                                                                                                 | These papers justify noise types such as incomplete skeletons, occlusion, and jitter, but they do not answer whether contextual ranking improves retrieval robustness.                                                                 |
| Skeleton learning           | metric   | SL-DML (Memmesheimer et al., 2020) and Skeleton-DML (Memmesheimer et al., 2022).                                                                                                                                                                                                         | These are the most direct baselines: they reduce one-shot skeleton action recognition to nearest-neighbor search in embedding space. They should be cited as prior art, not ignored.                                                   |
| Motion-language retrieval   | re-      | Text-to-Motion Retrieval (Petrovich et al., 2023), cross-dataset text-motion retrieval (Bensabath et al., 2024), tri-modal retrieval (Yin et al., 2024), multi-modal retrieval (Yu et al., 2025), and MoCHA (Warner et al., 2026).                                                       | This is where current motion retrieval is moving, but it changes the problem to text/video/audio-motion alignment. It is useful context and a later extension, not the first experiment.                                               |

GCN, Skeleton-DML/SL-DML, POEM/Pr-VIPE, text-to-motion retrieval, and MotionCLIP (Liao, nd; Zhu and collaborators, nd; OpenMMLab, nd; Yan, nd; Memmesheimer, nda,n; Google Research, nd; Petrovich et al., nd; Tevet and collaborators, nd). Tables 3 and 4 summarize the repository-level findings.

The most actionable observation is that MotionBERT’s one-shot NTU120 path already has three things the contextual loss needs: an embedding head, class-balanced sampling via PyTorch Metric Learning’s sampler, and cosine nearest-neighbor validation. A minimal first version can leave MotionBERT’s data code intact and swap only the loss and evaluator.

## 6 Experimental Design

The first experiment should isolate the effect of the loss. The proposed matrix is:

The training step is intentionally simple:

```

z = normalize(encoder(x))          # one embedding per motion
S = z @ z.T                      # cosine similarity matrix
Y = (labels[:, None] == labels[None]) # label agreement
C = contextual_similarity(S, k=M)  # M = samples per class
L = L_context(C, Y) + lambda * L_contrastive(z, labels)

```

The evaluator should be separate from the training loss. For each trained model, compute embeddings once for clean query/gallery sequences, then recompute query embeddings under each corruption level while keeping the gallery clean. This measures how retrieval degrades when the observed query pose is noisy, which is the use case motivating the project. A second variant can corrupt both query and gallery to simulate fully noisy pose-estimation archives.

Table 2: Dataset options for pose-sequence retrieval.

| Dataset                                                                           | Modality                                              | Scale / labels                                                                                         | Strength                                                                                                                                         | Weakness                                                                                                         | First? |
|-----------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------|
| NTU<br>RGB+D<br>120 (Liu<br>et al., 2020a)                                        | RGB, depth,<br>IR, 3D skele-<br>ton                   | 114,480 sam-<br>ples, 120 classes,<br>standard cross-<br>subject/cross-<br>setup splits                | Large class count<br>and multiple<br>examples per class<br>make label-based<br>retrieval well<br>defined; public<br>skeleton pipelines<br>exist. | Broad<br>daily/interpersonal<br>actions, not<br>fine-grained sports<br>technique.                                | Yes    |
| NTU<br>RGB+D<br>60                                                                | Same family                                           | 56,880 samples,<br>60 classes                                                                          | Easier and already<br>supported by<br>MotionBERT and<br>MMAAction2<br>configs.                                                                   | Smaller label space<br>than NTU120; less<br>aligned with<br>one-shot NTU120<br>metric-learning<br>papers.        | Pilot  |
| SportsPose<br>(Ingwersen<br>et al., 2023)                                         | Markerless 3D<br>sports pose                          | >176,000 poses,<br>24 subjects, 5<br>sports                                                            | Dynamic athletic<br>motion and sports<br>relevance; useful for<br>showing domain<br>relevance.                                                   | Pose-estimation<br>benchmark with<br>limited action<br>categories; retrieval<br>labels may need<br>construction. | Later  |
| AMASS+BABEL+SMPL/SMPL-<br>(Mahmood<br>et al., 2019;<br>Punnakkal<br>et al., 2021) | H mocap<br>with lan-<br>guage/action<br>labels        | AMASS has<br>40+ hours and<br>11k+ motions;<br>BABEL adds<br>28k+ sequence<br>and 63k+ frame<br>labels | Clean motion and<br>rich semantics;<br>good for general<br>motion retrieval.                                                                     | Label structure and<br>preprocessing are<br>more complex than<br>NTU; not the<br>fastest first result.           | Later  |
| FineGYM<br>(Shao et al.,<br>2020)                                                 | Video with<br>fine-grained<br>gymnastics<br>hierarchy | Fine-grained<br>temporal action<br>annotations                                                         | Strong for sports<br>phase retrieval and<br>hierarchical<br>relevance.                                                                           | Requires pose<br>extraction or use of<br>preprocessed<br>MMAAction2<br>skeleton<br>annotations.                  | Later  |
| H3WB<br>(Zhang et al.,<br>2022)                                                   | Whole-body<br>pose estima-<br>tion                    | 133 key-<br>points over<br>Human3.6M-<br>derived images                                                | Useful for whole-<br>body/keypoint<br>robustness.                                                                                                | Not naturally a<br>sequence retrieval<br>benchmark.                                                              | No     |
| UCF101<br>(Soomro<br>et al., 2012)                                                | Raw video                                             | 101 action<br>classes                                                                                  | Useful for eventual<br>video-to-pose<br>robustness.                                                                                              | Adds<br>pose-estimation<br>pipeline noise<br>before the retrieval<br>question is settled.                        | No     |

Table 3: Core implementation codebases.

| Codebase             | Reusable adaptation point                                                                                                                                                | Recommendation                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contextual loss repo | Hybrid contextual/contrastive loss, balanced sampler pattern, and cosine retrieval evaluators; replace the image loader with pose sequences and expose $k = M$ .         | Use as the loss/evaluator reference, not as the model scaffold.                                                                                           |
| MotionBERT           | NTU action configs, pretrained encoder, <code>get_representation</code> , normalized <code>ActionHeadEmbed</code> , PML sampler, and cosine nearest-neighbor validation. | Best first path: replace SupCon in <code>train_action_1shot.py</code> with contextual/hybrid loss and extend validation to Recall@K/mAP under corruption. |
| MMAction2 / PoseC3D  | Maintained skeleton toolbox, preprocessed NTU/FineGYM/UCF annotations, and ST-GCN/PoseC3D configs.                                                                       | Strong second-stage baseline: add an embedding head plus metric-learning loss/evaluator inside the MMAction2 stack.                                       |
| Standalone ST-GCN    | Simple model with explicit <code>extract_feature</code> ; remove the classifier head, pool sequence features, normalize, and train retrieval losses.                     | Useful sanity baseline, but the repo itself is historical.                                                                                                |

Table 4: Adjacent repositories and extension paths.

| Codebase                 | Reusable adaptation point                                                                 | Recommendation                                                                                                      |
|--------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Skeleton-DML / SL-DML    | Direct one-shot skeleton metric-learning framing and public code.                         | Closest baseline family; use if MotionBERT setup stalls or as prior-art comparison.                                 |
| POEM / Pr-UIPE           | View-invariant and occlusion-robust pose embedding code for cross-view pose retrieval.    | Valuable side experiment for single-frame or short-sequence robustness, not the first class-level motion benchmark. |
| Text-to-motion retrieval | Modular motion encoders, normalized text/motion embeddings, and Recall@K/NDCG evaluation. | Later extension only because it changes the problem to cross-modal retrieval.                                       |
| MotionCLIP               | AMASS/BABEL parsing and a CLIP-aligned latent motion space.                               | Useful for semantic motion search, but not core for the first pose-to-pose result.                                  |

Table 5: First-stage experimental matrix.

| Component   | Setting                                                                                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dataset     | NTU RGB+D 120 skeletons if available; otherwise NTU60 pilot using MotionBERT/MMAction2 preprocessed skeletons.                                                                                |
| Task        | Query/gallery action-class retrieval. A retrieved sequence is relevant if it shares the query’s action label; exclude the query instance when query and gallery are identical.                |
| Splits      | Cross-subject and cross-setup/cross-view where supported; report which split is used for each dataset.                                                                                        |
| Backbones   | MotionBERT embedding path first; then ST-GCN or PoseC3D through MMAction2. Optional simple TCN as a transparent lower-bound baseline.                                                         |
| Losses      | Cross-entropy penultimate embedding, triplet, SupCon, multi-similarity, contextual loss, and contextual+contrastive hybrid.                                                                   |
| Corruptions | Gaussian joint jitter, random joint dropout, limb dropout, confidence masking, temporal frame dropout, temporal jitter, scale/rotation/view perturbation, and left-right swap as an ablation. |
| Metrics     | Recall@1/5/10, mAP, mAP@R, clean-minus-noisy degradation, and area under the robustness curve across corruption severity.                                                                     |

## 7 Feasibility and Risks

Table 6: Feasibility tiers.

| Tier    | Path                                                                                  | Main risk                                                                                                               |
|---------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Low     | MotionBERT NTU one-shot embedding path plus contextual loss and synthetic corruption. | Loss/batch-size memory cost; contextual neighborhood quality depends on enough samples per class.                       |
| Medium  | MMAction2 ST-GCN/PoseC3D with custom embedding head and retrieval evaluator.          | Framework integration is heavier, but gives stronger baselines and preprocessed skeleton resources.                     |
| Higher  | SportsPose or FineGYM retrieval.                                                      | Labels and retrieval targets may need task-specific construction; pose extraction/preprocessing can become the project. |
| Highest | Rowing-specific or cross-dataset transfer.                                            | Data collection, labeling, domain shift, and biomechanical validity dominate the metric-learning question.              |

Compute risk is moderate. Skeleton sequences are much cheaper than raw video, and the first result can be run on a subset while debugging. The contextual objective scales with the batch similarity matrix, so memory pressure comes from balanced batch size. If needed, reduce sequence length, use MotionBERT-Lite, mixed precision, or gradient accumulation; do not reduce samples per class below the point where contextual neighborhoods become meaningless.

## 8 Conclusion

The concise conclusion for Prof. Kulis is that the proposed project is feasible and nontrivial if it is scoped correctly. Prior work has solved many pieces: contextual similarity for image retrieval, pose and skeleton encoders, one-shot skeleton metric learning, and robust skeleton action recogni-

tion. What remains underexplored is the controlled question of whether contextual neighborhoods improve temporal pose retrieval robustness under pose-estimation noise.

The recommended first deliverable is an NTU skeleton retrieval benchmark built from public code. Start with MotionBERT’s one-shot embedding pipeline, preserve its normalized embeddings and balanced sampler, add contextual/hybrid loss from the Kulis repository, and evaluate clean/noisy Recall@K and mAP. If this works, the natural second deliverable is MMAAction2 PoseC3D or ST-GCN++ under the same evaluator, followed by SportsPose or FineGYM only after the retrieval pipeline is stable.

## References

- Bensabath, L., Petrovich, M., and Varol, G. (2024). Cross-dataset human motion retrieval via proxies. *arXiv preprint arXiv:2405.16909*.
- Brown, A., Xie, W., Kalogeiton, V., and Zisserman, A. (2020). Smooth-ap: Smoothing the path towards large-scale image retrieval. In *European Conference on Computer Vision*, pages 677–694.
- Duan, H., Zhao, Y., Chen, K., Shao, D., Lin, D., and Dai, B. (2022). Revisiting skeleton-based action recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2969–2978.
- Google Research (n.d.). Poem. <https://github.com/google-research/google-research/tree/master/poem>. Accessed 2026-07-06.
- Hadsell, R., Chopra, S., and LeCun, Y. (2006). Dimensionality reduction by learning an invariant mapping. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 1735–1742.
- Ingwersen, C. K., Mikkelsen, C. M., Jensen, J. N., Hannemose, M. R., and Dahl, A. B. (2023). Sportspose: A dynamic 3d sports pose dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 5219–5228.
- Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., and Krishnan, D. (2020). Supervised contrastive learning. In *Advances in Neural Information Processing Systems*, volume 33, pages 18661–18673.
- Kulis, B. (n.d.). Brian kulis. <https://people.bu.edu/bkulis/>. Accessed 2026-07-06.
- Lee, I., Kim, D., Kang, S., and Lee, S. (2022). Mst-gcn: Multi-scale temporal graph convolutional network for skeleton-based action recognition. *arXiv preprint arXiv:2206.13028*.
- Liao, C. (n.d.). metric-learning-using-contextual-similarity. <https://github.com/Confusezius/metric-learning-using-contextual-similarity>. Accessed 2026-07-06.
- Liao, C., Tsiligkaridis, T., and Kulis, B. (2023). Supervised metric learning to rank for retrieval via contextual similarity optimization. In *Proceedings of the 40th International Conference on Machine Learning*, pages 21012–21033.
- Liu, J., Shahroudy, A., Perez, M., Wang, G., Duan, L.-Y., and Kot, A. C. (2020a). Ntu rgb+d 120: A large-scale benchmark for 3d human activity understanding. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(10):2684–2701.

- Liu, T., Sun, J. J., Zhao, L., Zhao, J., Yuan, L., Wang, Y., Chen, L.-C., Schroff, F., and Adam, H. (2020b). View-invariant, occlusion-robust probabilistic embedding for human pose. *arXiv preprint arXiv:2010.13321*.
- Mahmood, N., Ghorbani, N., Troje, N. F., Pons-Moll, G., and Black, M. J. (2019). Amass: Archive of motion capture as surface shapes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5442–5451.
- Memmesheimer, R. (n.d.a). skeleton-dml. <https://github.com/raphaelmemmesheimer/skeleton-dml>. Accessed 2026-07-06.
- Memmesheimer, R. (n.d.b). sl-dml. <https://github.com/raphaelmemmesheimer/sl-dml>. Accessed 2026-07-06.
- Memmesheimer, R., Haring, S., Theisen, N., and Paulus, D. (2022). Skeleton-dml: Deep metric learning for skeleton-based one-shot action recognition. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 3702–3710.
- Memmesheimer, R., Yang, H., Pfeiffer, S., and Paulus, D. (2020). Sl-dml: Signal-to-image representation and deep metric learning for skeleton-based action recognition. *arXiv preprint arXiv:2004.11085*.
- Mori, G., Pantofaru, C., Kothari, N., Leung, T., Toderici, G., Toshev, A., and Yang, W. (2015). Pose embeddings: A deep architecture for learning to match human poses. *arXiv preprint arXiv:1507.00302*.
- Musgrave, K., Belongie, S., and Lim, S.-N. (2020a). A metric learning reality check. *arXiv preprint arXiv:2003.08505*.
- Musgrave, K., Belongie, S., and Lim, S.-N. (2020b). Pytorch metric learning. *arXiv preprint arXiv:2008.09164*.
- OpenMMLab (n.d.). Mmaction2. <https://github.com/open-mmlab/mmaction2>. Accessed 2026-07-06.
- Petrovich, M., Black, M. J., and Varol, G. (2023). Text-to-motion retrieval: Towards joint understanding of human motion data and natural language. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2420–2425.
- Petrovich, M., Black, M. J., and Varol, G. (n.d.). text-to-motion-retrieval. <https://github.com/mesnico/text-to-motion-retrieval>. Accessed 2026-07-06.
- Punnakkal, A. R., Chandrasekaran, A., Athanasiou, N., Quiros-Ramirez, A., and Black, M. J. (2021). Babel: Bodies, action and behavior with english labels. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 722–731.
- Shao, D., Zhao, Y., Dai, B., and Lin, D. (2020). Finegym: A hierarchical video dataset for fine-grained action understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2616–2625.
- Shi, L., Zhang, Y., Cheng, J., and Lu, H. (2020). Pose refinement graph convolutional network for skeleton-based action recognition. *arXiv preprint arXiv:2010.07367*.



- Song, Y.-F., Zhang, Z., He, X., Cheng, J., and Lu, H. (2020). Richly activated graph convolutional network for robust skeleton-based action recognition. *arXiv preprint arXiv:2008.03791*.
- Soomro, K., Zamir, A. R., and Shah, M. (2012). Ucf101: A dataset of 101 human actions classes from videos in the wild. *arXiv preprint arXiv:1212.0402*.
- Sun, J. J., Zhao, J., Chen, L.-C., Schroff, F., Adam, H., and Liu, T. (2019). View-invariant probabilistic embedding for human pose. *arXiv preprint arXiv:1912.01001*.
- Tevet, G. and collaborators (n.d.). Motionclip. <https://github.com/GuyTevet/MotionCLIP>. Accessed 2026-07-06.
- Wang, X., Han, X., Huang, W., Dong, D., and Scott, M. R. (2019). Multi-similarity loss with general pair weighting for deep metric learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5022–5030.
- Warner, N., Taylor, C. E., Essa, I., and Sadhwani, A. (2026). Mocha: Enhancing text-motion retrieval with denoising caption supervision. *arXiv preprint arXiv:2603.23684*.
- Yan, S. (n.d.). St-gcn. <https://github.com/yysijie/st-gcn>. Accessed 2026-07-06.
- Yan, S., Xiong, Y., and Lin, D. (2018). Spatial temporal graph convolutional networks for skeleton-based action recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Yin, K., Zou, S., Ge, Y., and Tian, Z. (2024). Tri-modal motion retrieval by learning a joint embedding space. *arXiv preprint arXiv:2403.00691*.
- Yu, S., Wang, Z.-A., Yin, K., Tian, Z., Zhang, M., Si, W., and Zou, S. (2025). Multi-modal motion retrieval. *arXiv preprint arXiv:2507.23188*.
- Zhai, A. and Wu, H.-Y. (2018). Classification is a strong baseline for deep metric learning. *arXiv preprint arXiv:1811.12649*.
- Zhang, H., Tian, Y., Zhang, Y., Li, M., An, L., Sun, Z., and Liu, Y. (2022). Human3.6m whole-body: A benchmark for 3d whole-body human pose estimation. *arXiv preprint arXiv:2211.15672*.
- Zhu, W. and collaborators (n.d.). Motionbert. <https://github.com/Walter0807/MotionBERT>. Accessed 2026-07-06.
- Zhu, W., Ma, X., Liu, Z., Liu, L., Wu, W., and Wang, Y. (2023). Motionbert: A unified perspective on learning human motion representations. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 15085–15099.